BIO 005 · YUBA COLLEGE · FALL 2026

All 268 competencies, week by week.

Human Physiology · Section BIOL-5-D9286 · Sutter Internet (NET)

15 weeks · September 8 to December 16, 2026 · two exams

What this is. Every single thing this course will ask you to be able to do, sorted into the week it is taught. If it is not on this list, it is not on an exam. If it is on this list, it is fair game on the midterm that covers its week.

How to use it. Find your week. Those are your competencies. Your note sheet already has one page for each of them, so you do not have to copy anything off this list. Come back to it before a midterm and read down the five weeks it covers. Anything you cannot do without looking is your study list.

A note on the other packet. The Competency Packet in Canvas holds these same 268 grouped by unit, with the entry expectations in front. That one is for transfer and equivalency review. This one is for finding your week.

Dr. Sharilyn Rennie

MIDTERM 1 BLOCK

Weeks 1 to 7

119 competencies. Midterm 1. Covers Weeks 1 to 7. Window Monday October 26 at 8 am to Wednesday October 28 at 10 pm.

Week 1 How physiology works and what keeps you steady

Tue Sep 8 to Sun Sep 13 · 12 competencies

1 **Physiology and levels of function**

Define physiology and place a given process at the correct level of organization from molecule to organism, then state the level at which that process is best explained.

2 **Structure and function relationship** LAB

Predict how a change in the structure of a molecule, cell, tissue, or organ alters its function, using examples from two different organ systems.

3 **Homeostasis defined**

Define homeostasis, regulated variable, and setpoint, and distinguish homeostasis from chemical equilibrium and from steady state.

4 **Feedback loop components** LAB

Diagram a negative feedback loop labeling stimulus, sensor, afferent path, integrating center, efferent path, effector, and response, and trace body temperature or blood glucose through every step.

5 **Negative and positive feedback**

Distinguish negative from positive feedback by the direction of the response and give a physiological example of each, explaining why a positive feedback loop needs an outside event to end it.

6 **Feedforward control and acclimatization**

Explain anticipatory feedforward control and acclimatization and identify which one is operating in a given scenario.

7 **Local and reflex control pathways**

Distinguish a local control pathway from a long distance reflex pathway and classify a given response as one or the other.

8 **Mass balance** LAB

Apply the mass balance equation to a solute or to body water and calculate the intake, production, and output combination that holds the amount in the body constant.

9 **Units and unit conversion** LAB

Convert among the units used in physiology including molarity, osmolarity, milliequivalents, mmHg, liters per minute, and percent solutions.

10 **Graphing and data interpretation** LAB

Construct a labeled graph with the independent variable on the x axis, and read slope, direction, and trend from a physiological data set.

11 **Experimental design and controls** LAB

Identify the hypothesis, independent variable, dependent variable, and control condition in a physiology experiment and state what the control rules out.

12 **Measurement error and variability** LAB

Distinguish random from systematic error and explain why physiological measurements are repeated and averaged.

Week 2 The chemistry that does work in the body

Mon Sep 14 to Sun Sep 20 · 7 competencies

13 **Water and solution properties**

Explain how the polarity and hydrogen bonding of water determine solubility and identify whether a given solute is hydrophilic or hydrophobic.

14 **pH and buffers**

Define pH, state the normal pH range of arterial blood, and explain how a buffer pair resists a change in pH when acid or base is added.

15 **Protein structure and function**

Relate the levels of protein structure to binding site shape and explain how denaturation by heat or pH change destroys function.

16 **Enzyme activity and regulation**

Describe how enzymes lower activation energy and predict the effect of substrate concentration, temperature, pH, and competitive or allosteric inhibition on reaction rate.

17 **ATP and energy coupling**

Explain how ATP hydrolysis is coupled to endergonic cellular work and name three categories of work that require it.

18 **Enzyme assay**

Measure enzyme activity across a range of temperature or pH using a spectrophotometric or colorimetric assay, plot the results, and identify the optimum.

19 **ATP production pathways**

Compare glycolysis, the citric acid cycle, and oxidative phosphorylation by location, oxygen requirement, and ATP yield.

Week 3 Getting across the membrane

Mon Sep 21 to Sun Sep 27 · 16 competencies

20 **Body fluid compartments**

State the approximate volumes of total body water, intracellular fluid, extracellular fluid, plasma, and interstitial fluid in a 70 kg adult and compare the dominant solutes of the intracellular and extracellular compartments.

21 **Compartment separation and clinical volume shifts**

Predict the direction of water movement between compartments when extracellular osmolarity rises or falls and name a clinical situation that produces each shift.

22 **Membrane composition and fluidity**

Describe the fluid mosaic membrane and state how phospholipids, cholesterol, glycolipids, and integral and peripheral proteins contribute to its properties.

23 **Determinants of permeability**

Rank molecules by their ability to cross a lipid bilayer unaided using size, charge, and lipid solubility, and predict which will require a protein.

24 **Simple diffusion and Fick's law**

State the variables in Fick's law of diffusion and predict how a change in concentration gradient, surface area, membrane thickness, or distance changes the rate of transfer.

25 **Osmolarity and tonicity**

Calculate the osmolarity of a solution, distinguish osmolarity from tonicity, and classify a solution as isotonic, hypotonic, or hypertonic to a cell.

26 **Osmosis and cell volume**

Predict the direction of water movement and the resulting change in cell volume when a cell is placed in a solution of stated osmolarity and penetrating solute content.

27 **Diffusion and osmosis experiment**

Measure diffusion and osmotic movement across a selectively permeable membrane and relate the observed rate to molecular size and concentration gradient.

28 **Tonicity and red blood cells**

Observe erythrocytes in solutions of different tonicity, identify crenation, normal shape, and hemolysis, and explain each result.

29 **Facilitated diffusion**

Describe carrier and channel mediated diffusion and explain why carrier mediated transport shows saturation and specificity while simple diffusion does not.

30 **Primary active transport**

Explain how the sodium potassium ATPase uses ATP to move three sodium out and two potassium in, and state the two gradients it maintains.

31 **Secondary active transport**

Distinguish symport from antiport and trace how the sodium gradient powers glucose uptake by SGLT and calcium removal by the sodium calcium exchanger.

32 **Transport maximum and saturation**

Interpret a transport rate curve, identify the transport maximum, and apply the concept to renal glucose handling in hyperglycemia.

33 **Vesicular transport**

Compare phagocytosis, pinocytosis, receptor mediated endocytosis, and exocytosis by trigger, cargo, and energy requirement.

34 **Transepithelial transport**

Trace glucose or sodium from lumen to blood across a polarized epithelium and identify which step occurs at the apical membrane and which at the basolateral membrane.

35 **Transport simulation**

Use a membrane transport simulation to distinguish simple diffusion from facilitated diffusion and active transport by their response to gradient reversal and metabolic poison.

Week 4 How cells talk, and the electrical signal

Mon Sep 28 to Sun Oct 4 · 20 competencies

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36 **Ion distribution and electrochemical gradients**

State the typical intracellular and extracellular concentrations of sodium, potassium, chloride, and calcium and separate the chemical from the electrical component of the driving force on each ion.

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37 **Nernst equation**

Calculate the equilibrium potential for an ion with the Nernst equation and explain what the sign of the result means for the direction that ion will move.

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38 **Resting membrane potential**

Explain why the resting membrane potential sits near the potassium equilibrium potential and predict how it shifts when membrane permeability to potassium or sodium changes.

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39 **Ion channel gating**

Compare leak, voltage gated, ligand gated, and mechanically gated channels by what opens each one and give a physiological location for each.

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40 **Depolarization and hyperpolarization**

Define depolarization, repolarization, hyperpolarization, and overshoot and label each on a membrane potential tracing.

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41 **Membrane potential simulation**

Manipulate extracellular potassium and sodium in a simulation and record the resulting change in resting membrane potential against the Nernst prediction.

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42 **Neuron structural and functional classes**

Classify a neuron as multipolar, bipolar, or pseudounipolar by structure and as sensory, motor, or interneuron by function, and match each class to a location in the nervous system.

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43 **Functional regions of a neuron**

Label the dendrites, cell body, axon hillock, trigger zone, axon, and axon terminal and state which signal type each region carries.

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44 **Glial cell functions**

Match astrocytes, oligodendrocytes, microglia, ependymal cells, satellite cells, and Schwann cells to their physiological roles.

45 **Myelin and its loss**

Explain how myelin and the nodes of Ranvier speed conduction and predict the functional consequence of demyelination.

46 **Axonal transport**

Distinguish fast anterograde and retrograde axonal transport from slow transport and state what each carries.

47 **Graded potentials**

Describe how a graded potential is produced and explain why it varies with stimulus strength and decays with distance, using the terms current leak and cytoplasmic resistance.

48 **Action potential phases**

Diagram an action potential and state the channel state and ion movement responsible for depolarization to peak and for repolarization and afterhyperpolarization.

49 **Threshold and all or none**

Explain what threshold represents at the trigger zone and why action potential amplitude does not change with a stronger stimulus.

50 **Coding of stimulus intensity**

Explain how the nervous system encodes stimulus strength when action potential amplitude is fixed.

51 **Refractory periods**

Distinguish the absolute from the relative refractory period by channel state and explain how the refractory period sets maximum firing frequency and prevents backward conduction.

52 **Conduction velocity**

Rank axons by conduction velocity using diameter and myelination and contrast continuous with saltatory conduction.

53 **Effects of altered extracellular ions**

Predict the effect of hyperkalemia and hypokalemia on resting potential and excitability and explain why local anesthetics that block sodium channels abolish the action potential.

54 **Action potential simulation**

Run a neuron simulation with sodium and potassium channel blockers and interpret the resulting change in the action potential tracing.

55 **Nerve conduction measurement**

Measure conduction velocity from a recorded compound action potential and account for the difference between the fastest and slowest fibers.

Week 5 Synapses and central integration

Mon Oct 5 to Sun Oct 11 · 16 competencies

56 **Sequence at a chemical synapse**

Order the events of chemical synaptic transmission from action potential arrival through calcium entry and vesicle fusion to receptor binding and postsynaptic response.

57 **Neurotransmitter classes**

Match acetylcholine, the catecholamines, serotonin, glutamate, GABA, glycine, and the neuropeptides to their usual excitatory or inhibitory effect and to a site of action.

58 **Neurotransmitter removal**

Name the three routes that clear a neurotransmitter from the synaptic cleft, degradation, reuptake, and diffusion, and give a drug that blocks one of them.

59 **Excitatory and inhibitory postsynaptic potentials**

Distinguish an EPSP from an IPSP by the ion channel opened and the direction of the membrane potential change.

60 **Summation and integration**

Distinguish temporal from spatial summation and determine whether a stated combination of EPSPs and IPSPs will bring the trigger zone to threshold.

61 **Presynaptic modulation**

Compare presynaptic inhibition and facilitation with postsynaptic modulation and state the advantage of presynaptic control at a single input, which is selectivity.

62 **Synaptic plasticity**

Explain long term potentiation as a mechanism of learning and identify the roles of repeated stimulation and receptor insertion.

63 **Electrical synapses**

Compare an electrical synapse at a gap junction with a chemical synapse by speed and by capacity for modulation.

64 **Synapse simulation**

Alter neurotransmitter release and receptor availability in a synapse simulation and interpret the resulting postsynaptic recording.

65 **Reflex arc components**

Diagram a reflex arc naming the receptor, sensory neuron, integrating center, motor neuron, and effector, and classify the reflex as monosynaptic or polysynaptic.

66 **Muscle spindle and stretch reflex**

Explain how the muscle spindle detects stretch and trace the stretch reflex including reciprocal inhibition of the antagonist.

67 **Golgi tendon organ**

Contrast the stimulus and the reflex response of the Golgi tendon organ with those of the muscle spindle.

68 **Withdrawal and crossed extensor reflexes**

Trace the flexor withdrawal reflex and the crossed extensor reflex and explain how the two together preserve balance.

69 **Ascending and descending pathways**

Compare the dorsal column, spinothalamic, and corticospinal pathways by the information carried, the side of decussation, and the deficit produced by a lesion.

70 **Cerebrospinal fluid and the blood brain barrier**

State the functions of cerebrospinal fluid and explain how the blood brain barrier selects what reaches neurons, including why lipid soluble drugs cross readily.

71 **Reflex testing and reaction time**

Elicit and grade deep tendon reflexes and measure reaction time to distinguish a reflex response from a voluntary response by latency.

Week 6 Sensing the world, and the responses you do not control

Mon Oct 12 to Sun Oct 18 · 24 competencies

72 **Sensory transduction**

Explain how a stimulus is converted into a receptor potential and how that graded potential becomes a train of action potentials in the sensory neuron.

73 **Receptor classification**

Classify a receptor by stimulus type as a mechanoreceptor, chemoreceptor, thermoreceptor, photoreceptor, or nociceptor and give a body location for each.

74 **Stimulus coding**

Explain how the nervous system encodes modality, location, intensity, and duration and identify the labeled line principle.

75 **Receptive fields and acuity**

Relate receptive field size and lateral inhibition to two point discrimination and predict which body regions have the finest acuity.

76 **Receptor adaptation**

Distinguish tonic from phasic receptors by their adaptation rate and match each to a stimulus the body needs to monitor continuously or only at onset.

77 **Somatosensory pathways**

Trace touch and pain information from receptor to cortex and explain why the somatosensory homunculus is disproportionate.

78 **Pain and its modulation**

Distinguish fast and slow pain by fiber type, explain referred pain by convergence, and describe how gate control and endogenous opioids reduce pain transmission.

79 **Tactile mapping and adaptation testing**

Measure two point discrimination at several body sites and show receptor adaptation and referred sensation experimentally.

80 **Optics and accommodation**

Explain refraction by the cornea and lens and describe how accommodation and the pupillary reflex focus near and far images.

81 **Phototransduction**

Trace phototransduction from photon absorption by retinal through the change in cyclic GMP to the decrease in glutamate release and explain why photoreceptors hyperpolarize to light.

82 **Rods cones and visual processing**

Compare rods and cones by sensitivity and acuity and trace the visual pathway to predict the field deficit produced by a lesion at the optic nerve or the optic chiasm.

83 **Refractive errors and clinical vision**

Explain myopia, hyperopia, presbyopia, and astigmatism in terms of focal point and eyeball or lens shape and state the corrective lens for each.

84 **Auditory transduction**

Trace sound from the tympanic membrane through the ossicles and cochlear fluid to hair cell bending and auditory nerve firing and explain how pitch and loudness are coded.

85 **Conductive and sensorineural hearing loss**

Distinguish conductive from sensorineural hearing loss by the site of the lesion and by the expected Weber and Rinne results.

86 **Equilibrium**

Explain how the semicircular canals detect angular acceleration and how the utricle and saccule detect linear acceleration and head position.

87 **Taste and smell**

Compare gustatory and olfactory transduction and explain why olfactory input reaches the limbic system without a thalamic relay.

88 **Vision testing**

Perform visual acuity, blind spot, accommodation, and color vision testing and interpret each result.

89 **Hearing and equilibrium testing**

Perform Weber and Rinne tuning fork tests and a balance test and interpret the findings.

90 **Autonomic and somatic organization**

Contrast the somatic motor system with the autonomic two neuron pathway by neuron count and by effector tissue.

91 **Sympathetic and parasympathetic divisions**

Compare the two autonomic divisions by CNS origin and ganglion location and predict the effect of each on heart rate and on the gastrointestinal tract.

92 **Autonomic neurotransmitters and receptors**

Match nicotinic, muscarinic, alpha, and beta receptors to their neurotransmitter, their location, and the response each produces.

93 **Dual innervation and tonic control**

Explain dual innervation and autonomic tone and predict the effect of removing one division from a dually innervated organ.

94 **Autonomic pharmacology**

Predict the effect of a beta blocker or a muscarinic antagonist on a named organ from the receptor it targets.

95 **Autonomic function testing**

Measure heart rate variability and the response to a cold pressor or orthostatic challenge and interpret the result as sympathetic or parasympathetic dominance.

Week 7 Muscle, and how movement gets commanded

Mon Oct 19 to Sun Oct 25 · 24 competencies

96 **Neuromuscular junction**

Trace transmission at the neuromuscular junction from the motor neuron action potential to the end plate potential and explain why every action potential in the motor neuron produces one in the muscle fiber.

97 **Excitation contraction coupling**

Trace excitation contraction coupling from the T tubule through the DHP and ryanodine receptors to calcium release from the sarcoplasmic reticulum.

98 **Crossbridge cycle**

Order the steps of the crossbridge cycle, binding, power stroke, detachment, and reactivation, and state where ATP binds and where it is hydrolyzed.

99 **Calcium and the regulatory proteins**

Explain how calcium binding to troponin moves tropomyosin off the myosin binding site and predict what happens to contraction if calcium cannot be released.

100 **Relaxation and calcium removal**

Explain how SERCA and acetylcholinesterase end a contraction and predict the result if either one fails.

101 **The muscle twitch**

Label the latent period, contraction phase, and relaxation phase on a twitch tracing and explain what limits each phase.

102 **Summation and tetanus**

Explain wave summation and distinguish unfused from fused tetanus by stimulus frequency and calcium accumulation.

103 **Motor units and recruitment**

Define the motor unit and explain how size of the unit and orderly recruitment grade the force a whole muscle produces.

104 **Length tension relationship**

Interpret a length tension curve and explain why tension falls at very short and very long sarcomere lengths.

105 **Isometric and isotonic contraction**

Distinguish isometric, concentric, and eccentric contractions and predict shortening velocity from the load applied.

106 **Muscle fiber types**

Compare slow oxidative, fast oxidative glycolytic, and fast glycolytic fibers by speed, ATP source, fatigue resistance, and typical task.

107 **ATP sources during activity**

Order creatine phosphate, anaerobic glycolysis, and oxidative phosphorylation by how quickly each supplies ATP and how long each can sustain it.

108 **Fatigue and oxygen debt**

Distinguish central from peripheral fatigue and explain the metabolic basis of excess post exercise oxygen consumption.

109 **Muscle plasticity**

Predict the change in fiber size, capillary supply, and mitochondrial content produced by endurance training, resistance training, and disuse.

110 **Electromyography and grip fatigue**

Record grip force and surface EMG over a sustained contraction and relate the decline in force to recruitment and fatigue.

111 **Twitch and tetanus simulation**

Generate twitch and tetanus tracings in a muscle simulation by varying stimulus voltage and frequency and identify threshold and maximal stimulus.

112 **Smooth muscle contraction mechanism**

Trace smooth muscle contraction through calcium entry and release to calmodulin and myosin light chain kinase and explain why the mechanism is slower and more economical than in skeletal muscle.

113 **Smooth muscle types and regulation**

Compare single unit and multi unit smooth muscle and list the stimuli that alter smooth muscle tone including stretch and local chemical signals.

114 **Cardiac muscle properties**

Explain how intercalated discs and gap junctions allow the myocardium to act as a functional syncytium and state why cardiac muscle cannot be tetanized.

115 **Comparison of the three muscle types**

Build a comparison of skeletal, cardiac, and smooth muscle by calcium source, regulatory protein, control, and speed of contraction.

116 **Motor control hierarchy**

Order the levels of motor control from spinal reflex through brainstem to cortex and state what each level contributes to a voluntary movement.

117 **Corticospinal pathway**

Trace the corticospinal pathway from the primary motor cortex to the skeletal muscle and identify where it crosses.

118 **Cerebellum and basal ganglia**

Contrast the contributions of the cerebellum and the basal ganglia to movement and match a described motor sign to the structure involved.

119 **Upper and lower motor neuron signs**

Distinguish upper from lower motor neuron lesions by tone, reflexes, and muscle bulk and assign a set of findings to the correct level.

MIDTERM 2 BLOCK

Weeks 8 to 14

142 competencies. Midterm 2. Covers Weeks 8 to 14. Window Monday December 14 at 8 am to Wednesday December 16 at 10 pm.

Week 8 Hormones and reproduction, the slow control system

Mon Oct 26 to Sun Nov 1 · 25 competencies

120 **Signal types and range**
Classify a chemical signal as autocrine, paracrine, neurotransmitter, neurohormone, or hormone by its route and distance of travel.

121 **Receptor location and ligand solubility**
Predict whether a signal molecule binds a surface receptor or an intracellular receptor from its lipid solubility and relate that to the speed and duration of the response.

122 **G protein coupled receptors**
Trace a G protein coupled receptor pathway from ligand binding through the G protein and amplifier enzyme to the second messenger and the cellular response.

123 **Second messengers**
Identify cyclic AMP, IP3, diacylglycerol, and calcium as second messengers and state the enzyme that generates each and one target it activates.

124 **Catalytic receptors and intracellular receptors**
Compare a receptor enzyme such as the insulin receptor with an intracellular steroid receptor by mechanism and by time course of the response.

125 **Signal amplification**
Explain how a cascade amplifies a signal and estimate the size of the amplification across the steps of a given pathway.

126 **Receptor modulation**
Define agonist, antagonist, competitive inhibition, up regulation, and down regulation and predict the effect of chronic ligand excess on target cell sensitivity.

127 **Signal termination**
Name three mechanisms that end a chemical signal and explain why a signal that cannot be terminated produces pathology.

128 **Dose response relationships**

Plot a dose response curve, identify threshold, maximal response, and EC50, and compare a full agonist with a partial agonist on the same axes.

129 **Hormone classes**

Classify hormones as peptide, steroid, or amine and compare the three by synthesis, storage, and receptor location.

130 **Hormone transport and half life**

Explain how protein binding in plasma affects the free hormone fraction and the half life and predict which hormone class circulates bound.

131 **Hormone receptors and target response**

Explain why one hormone can produce different responses in different tissues and relate receptor number to target cell sensitivity.

132 **Hormone interactions**

Define synergism, permissiveness, and antagonism and identify each in a described hormone pair.

133 **Control of hormone release**

Classify a hormone stimulus as humoral, neural, or hormonal and trace an example of each.

134 **Hypothalamic pituitary axes**

Trace a three tier axis from hypothalamic releasing hormone through the anterior pituitary trophic hormone to the peripheral gland and place the long and short loop negative feedback.

135 **Primary and secondary endocrine disorders**

Use hormone levels at two tiers of an axis to classify a disorder as primary or secondary and as hypersecretion or hyposecretion.

136 **Posterior pituitary hormones**

Explain why the posterior pituitary is neural tissue and state the stimulus and target action of antidiuretic hormone and oxytocin.

137 **Hormone assay**

Run or simulate an immunoassay for a hormone and use the standard curve to determine an unknown concentration.

138 **Hypothalamic pituitary gonadal axis**

Trace the gonadotropin releasing hormone axis to the gonads and place the feedback loops for both sexes.

139 **Male reproductive physiology**

Explain the control of spermatogenesis by follicle stimulating hormone, luteinizing hormone, and testosterone and state the roles of the Sertoli and Leydig cells.

140 **Ovarian cycle**

Order the follicular, ovulatory, and luteal phases and explain how the luteinizing hormone surge is triggered by positive feedback.

141 **Uterine cycle**

Correlate the menstrual, proliferative, and secretory phases of the uterine cycle with the estrogen and progesterone levels that drive them.

142 **Pregnancy and placental hormones**

Explain how human chorionic gonadotropin rescues the corpus luteum and state the roles of placental estrogen, progesterone, and relaxin.

143 **Parturition and lactation**

Explain labor as a positive feedback loop driven by oxytocin and compare the hormonal control of milk production with milk ejection.

144 **Hormone cycle graph interpretation**

Read a graph of gonadotropin and ovarian hormone levels across a cycle and identify the day of ovulation and the event driving each peak.

Week 9 The heart as a pump

Mon Nov 2 to Sun Nov 8 · 17 competencies

145 **Adrenal medulla**

Explain why the adrenal medulla is a modified sympathetic ganglion and compare the duration of its circulating catecholamine effect with direct sympathetic innervation.

146 **Pacemaker potential**

Explain the unstable pacemaker potential of the sinoatrial node by the funny current, calcium entry, and potassium exit and state what sets intrinsic heart rate.

147 **Contractile cell action potential**

Diagram the ventricular action potential and identify the ion movement responsible for the rapid upstroke, the plateau, and repolarization.

148 **Conduction system**

Trace an impulse through the sinoatrial node, internodal pathways, atrioventricular node, bundle of His, bundle branches, and Purkinje fibers and explain the purpose of the atrioventricular delay.

149 **Cardiac refractory period**

Explain why the long refractory period of cardiac muscle prevents tetanus and why that matters for pump function.

150 **ECG waves and intervals**

Label the P wave, QRS complex, T wave, PR interval, and QT interval and state the electrical event each represents.

151 **ECG interpretation**

Calculate heart rate from an ECG strip and identify sinus rhythm, tachycardia, bradycardia, and a first degree or complete heart block.

152 **ECG recording**

Record a lead II ECG, measure the intervals, and correlate each wave with the mechanical event that follows it.

153 **Cardiac cycle**

Order the phases of the cardiac cycle and state the pressure relationship that opens and closes the atrioventricular and semilunar valves in each phase.

154 **Pressure volume loop**

Label the four phases of a ventricular pressure volume loop and predict how the loop changes with increased preload, increased afterload, or increased contractility.

155 **Heart sounds**

Explain the origin of the first and second heart sounds and relate a murmur to valve stenosis or regurgitation by its timing.

156 **Stroke volume and ejection fraction**

Calculate stroke volume and ejection fraction from end diastolic and end systolic volumes and interpret a reduced ejection fraction.

157 **Cardiac output**

Calculate cardiac output and cardiac reserve and predict the effect of a change in heart rate or stroke volume on each.

158 **Frank Starling relationship**

State the Frank Starling law and explain how venous return and end diastolic volume determine the force of the next contraction.

159 **Preload afterload and contractility**

Distinguish preload, afterload, and contractility and predict how a change in each alters stroke volume.

160 **Autonomic regulation of the heart**

Explain how sympathetic and parasympathetic input alter heart rate and contractility at the receptor and second messenger level.

161 **Heart sound and pulse correlation**

Auscultate the heart sounds, locate the valve areas, and correlate the sounds with the pulse and with the cardiac cycle phases.

Week 10 Pressure, flow, and holding blood pressure steady

Mon Nov 9 to Sun Nov 15 · 13 competencies

162 **Adrenal cortex**

Match the three cortical zones to aldosterone, cortisol, and the adrenal androgens and state the primary action of each.

163 **Pressure flow and resistance**

Apply the relationship among flow, pressure gradient, and resistance and predict the effect of a change in vessel radius, blood viscosity, or vessel length on flow.

164 **Vessel structure and function**

Match arteries, arterioles, capillaries, venules, and veins to their role in pressure buffering, resistance, exchange, and capacitance.

165 **Arterial blood pressure**

Define systolic, diastolic, pulse, and mean arterial pressure, calculate mean arterial pressure, and explain how it is determined by cardiac output and total peripheral resistance.

166 **Local control of blood flow**

Explain myogenic, metabolic, and endothelial control of arteriolar tone and predict the local flow response to increased tissue metabolism.

167 **Capillary exchange and Starling forces**

Identify the four Starling forces at a capillary and determine the net direction of fluid movement at the arterial and venous ends.

168 **Edema**

Explain how increased capillary hydrostatic pressure, reduced plasma protein, increased permeability, or lymphatic blockage each produce edema and match a clinical scenario to its mechanism.

169 **Venous return**

Explain the skeletal muscle pump, the respiratory pump, and venoconstriction and predict the effect of prolonged standing on venous return and stroke volume.

170 **Blood pressure measurement**

Measure blood pressure by auscultation, explain the origin of the Korotkoff sounds, and record the response to a postural or exercise challenge.

171 **Baroreceptor reflex**

Trace the baroreceptor reflex from the carotid sinus and aortic arch through the medulla to the heart and vessels and state the response to a fall in blood pressure.

172 **Hormonal control of blood pressure**

Describe the roles of angiotensin II, aldosterone, antidiuretic hormone, and atrial natriuretic peptide in long term blood pressure control.

173 **Cardiovascular response to exercise**

Predict the change in heart rate, stroke volume, cardiac output, total peripheral resistance, and regional blood flow distribution during moderate exercise.

174 **Compensation in hemorrhage and shock**

Sequence the compensatory responses to blood loss and explain the point at which compensation fails.

Week 11 Blood and how the body defends itself

Mon Nov 16 to Sun Nov 22 · 19 competencies

175 **Plasma and blood functions**

State the three functional categories of blood, transport, regulation, and protection, and name the major plasma proteins and the function of each.

176 **Hematocrit and formed elements**

Define hematocrit and interpret a value as normal or abnormal in the context of anemia and dehydration.

177 **Erythrocyte structure and hemoglobin**

Relate the biconcave shape and absent nucleus of the erythrocyte to its function and describe how the four heme groups of hemoglobin carry oxygen.

178 **Erythropoiesis and its regulation**

Trace the erythropoietin negative feedback loop from tissue hypoxia to increased red cell production and state the nutrients required.

179 **Red cell destruction and bilirubin**

Trace the fate of hemoglobin after red cell breakdown and explain how failure to clear bilirubin produces jaundice.

180 **Leukocytes**

Match neutrophils, lymphocytes, monocytes, eosinophils, and basophils to their defensive function and to the condition that raises each.

181 **Hemostasis**

Order the three phases of hemostasis, vascular spasm, platelet plug formation, and coagulation, and identify the role of von Willebrand factor and thromboxane.

182 **Coagulation cascade**

Compare the intrinsic and extrinsic pathways to the common pathway and identify the roles of thrombin, fibrin, calcium, and vitamin K.

183 **Clot limitation and fibrinolysis**

Explain how anticoagulants and plasmin limit and remove a clot and predict the effect of a drug that blocks vitamin K or platelet aggregation.

184 **ABO and Rh blood types**

Predict agglutination for any donor and recipient ABO and Rh combination and explain the mechanism of hemolytic disease of the newborn.

185 **Hematocrit and hemoglobin determination**

Determine hematocrit and hemoglobin from a sample or simulation and evaluate the result against reference ranges.

186 **Blood typing**

Perform simulated ABO and Rh typing and determine compatible donor types for the result obtained.

187 **Lymphatic return**

State the volume of fluid the lymphatic system returns each day and trace lymph from an interstitial space to the subclavian vein.

188 **Innate defenses**

Describe the physical, chemical, and cellular innate defenses and explain what makes them nonspecific and immediate.

189 **Inflammation and fever**

Order the events of the inflammatory response and explain how the cardinal signs arise and why fever aids defense.

190 **Adaptive immunity**

Compare humoral and cell mediated immunity by the cell responsible, the target, and the mechanism of elimination.

191 **Antibody structure and function**

Relate antibody structure to antigen binding and name the mechanisms by which antibodies neutralize or mark a target.

192 **Immune memory and immunization**

Compare the primary and secondary antibody responses and explain how active and passive immunization produce protection.

193 **Immune dysfunction**

Classify allergy, autoimmunity, and immunodeficiency by what the immune system is doing wrong in each.

Week 12 Digestion, and how you use food for fuel

Mon Nov 23 to Sun Nov 29, Thanksgiving week · 23 competencies

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194 **Growth hormone**

Describe the direct and insulin like growth factor mediated actions of growth hormone and predict the result of excess or deficiency before and after epiphyseal closure.

195 **Thyroid hormone**

Trace thyroid hormone synthesis, state the metabolic actions of T3 and T4, and match hyperthyroid and hypothyroid signs to the underlying rate of metabolism.

196 **Cortisol and the stress response**

Describe the metabolic, immune, and cardiovascular actions of cortisol and explain the consequences of chronic elevation.

197 **Pancreatic islet hormones**

Compare insulin and glucagon by stimulus and by effect on liver glycogen and on plasma glucose and identify the islet cell that secretes each.

198 **Diabetes mellitus**

Distinguish type 1 from type 2 diabetes by mechanism and explain how untreated hyperglycemia produces polyuria and ketoacidosis.

199 **Glucose tolerance testing**

Plot a glucose tolerance curve from measured or simulated data and distinguish a normal response from an impaired one.

200 **Four digestive processes**

Distinguish motility, secretion, digestion, and absorption and identify where in the tract each dominates.

201 **Motility patterns**

Compare peristalsis, segmentation, and the migrating motor complex by pattern and purpose.

202 **Neural and hormonal regulation**

Explain short and long reflex control by the enteric nervous system and match gastrin, secretin, cholecystokinin, and glucose dependent insulinotropic peptide to their stimulus and action.

203 **Phases of digestive control**

Describe the cephalic, gastric, and intestinal phases and state the dominant control mechanism in each.

204 **Gastric secretion and mucosal protection**

Trace the production of hydrochloric acid by the parietal cell and explain how the mucosal barrier protects the stomach and how ulcers develop when it fails.

205 **Pancreatic and biliary secretion**

State the components of pancreatic juice and bile and explain how bicarbonate neutralizes chyme and how bile salts emulsify fat.

206 **Carbohydrate and protein digestion and absorption**

Trace a starch and a protein from mouth to bloodstream naming the enzymes at each step and the transport mechanism at the brush border.

207 **Lipid digestion and absorption**

Trace a triglyceride through emulsification, lipase action, micelle formation, and chylomicron packaging to the lymphatic route and explain why fat takes a different path than glucose.

208 **Liver function and enterohepatic circulation**

State the metabolic, storage, and detoxification functions of the liver and trace the enterohepatic circulation of bile salts.

209 **Large intestine and microbiome**

Describe water and electrolyte absorption in the colon, the contribution of the microbiome, and the defecation reflex.

210 **Digestive enzyme experiment**

Test the effect of pH, temperature, and substrate on a digestive enzyme and relate the results to the region of the tract where that enzyme works.

211 **Absorptive state**

Describe the fate of glucose, amino acids, and fats in the absorptive state and identify insulin as the dominant hormone.

212 **Postabsorptive state**

Explain how glycogenolysis, gluconeogenesis, lipolysis, and ketogenesis maintain plasma glucose during fasting and name the hormones that drive each.

213 **Integrated glucose regulation**

Predict the hormonal and metabolic response to a carbohydrate meal and to a prolonged fast and explain how plasma glucose stays within range in both.

214 **Energy balance and metabolic rate**

Define basal metabolic rate, state the factors that raise and lower it, and explain how indirect calorimetry estimates energy expenditure.

215 **Thermoregulation**

Trace the negative feedback response to heat and to cold through the hypothalamus and the effectors and explain the mechanism of fever.

216 **Metabolic rate measurement**

Estimate metabolic rate from measured or simulated oxygen consumption and compare resting values with values after activity.

Week 13 Breathing, gas transport, and the fast pH lever

Mon Nov 30 to Sun Dec 6 · 24 competencies

217 **Functions and functional zones**

State the functions of the respiratory system and distinguish the conducting zone from the respiratory zone by structure and by role in gas exchange.

218 **Pressure gradients and airflow**

Apply Boyle's law to inspiration and expiration and relate alveolar pressure to atmospheric pressure at each point in a quiet breath.

219 **Intrapleural pressure**

Explain why intrapleural pressure is subatmospheric and predict what happens to the lung and chest wall when the pleural seal is broken.

220 **Compliance and elastic recoil**

Define lung compliance and elastic recoil and predict the effect of fibrosis and of emphysema on each.

221 **Surfactant and surface tension**

Explain how surfactant lowers surface tension and why its absence causes alveolar collapse in the premature newborn.

222 **Airway resistance**

Identify the main determinant of airway resistance and predict the effect of bronchoconstriction, mucus, and sympathetic stimulation on airflow.

223 **Lung volumes and capacities**

Define tidal volume, inspiratory and expiratory reserve volume, residual volume, vital capacity, and total lung capacity and read each from a spirogram.

224 **Obstructive and restrictive patterns**

Use the ratio of forced expiratory volume in one second to forced vital capacity to classify a spirometry result as obstructive or restrictive.

225 **Alveolar ventilation and dead space**

Calculate minute ventilation and alveolar ventilation and explain why slow deep breathing ventilates the alveoli better than rapid shallow breathing.

226 **Spirometry**

Record a spirogram, calculate the lung volumes and capacities available from it, and compare the results with predicted values.

227 **Partial pressures**

Apply Dalton's law to calculate the partial pressure of oxygen in inspired and alveolar air and state the normal partial pressures in alveolar air, arterial blood, and venous blood.

228 **Diffusion at the respiratory membrane**

Apply the determinants of diffusion to the respiratory membrane and predict the effect of edema, fibrosis, and emphysema on gas transfer.

229 **Ventilation perfusion matching**

Explain how local hypoxic vasoconstriction and bronchiolar responses match ventilation to perfusion and predict the consequence of a mismatch.

230 **Oxygen transport and the oxyhemoglobin curve**

Explain the sigmoid shape of the oxyhemoglobin dissociation curve and state the physiological advantage of the steep and the flat portions.

231 **Curve shifts**

Predict the direction of the oxyhemoglobin curve shift produced by a change in pH, carbon dioxide, temperature, or 2,3 bisphosphoglycerate and state what the shift means for tissue oxygen delivery.

232 **Carbon dioxide transport**

Name the three forms in which carbon dioxide is carried, state the approximate percentage of each, and trace the chloride shift in a systemic capillary.

233 **Bohr and Haldane effects**

Distinguish the Bohr effect from the Haldane effect and explain how the two work together at the tissue and at the lung.

234 **Oxygen content versus partial pressure**

Distinguish oxygen content from oxygen partial pressure and saturation and explain why anemia and carbon monoxide poisoning reduce delivery differently.

235 **Respiratory centers**

Identify the medullary and pontine respiratory centers and explain how the basic breathing rhythm is generated and modified.

236 **Chemoreceptor control**

Compare central and peripheral chemoreceptors by stimulus and explain why carbon dioxide is the primary minute to minute drive to breathe.

237 **Ventilation in exercise and at altitude**

Predict the ventilatory response to exercise and to acute and chronic altitude exposure and explain the acclimatization changes.

238 **Ventilatory response testing**

Measure the change in breathing rate and depth after hyperventilation, breath holding, and rebreathing and explain each result by the change in carbon dioxide.

239 **Buffer systems**

Compare the bicarbonate, phosphate, and protein buffer systems by location and speed and explain why bicarbonate is the dominant extracellular buffer.

240 **Respiratory control of pH**

Explain how a change in ventilation shifts the carbon dioxide and bicarbonate equilibrium and therefore plasma pH.

Week 14 The kidney and body fluid balance

Mon Dec 7 to Sun Dec 13 · 21 competencies

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241 **Calcium homeostasis**

Trace the response to falling plasma calcium through parathyroid hormone, calcitriol, and the actions on bone, kidney, and intestine, and state the role of calcitonin.

242 **Kidney functions**

List the homeostatic functions of the kidney including water and electrolyte balance, acid base regulation, waste excretion, blood pressure control, and hormone production.

243 **Nephron structure and function**

Match each nephron segment to its dominant transport function and distinguish a cortical from a juxtamedullary nephron by capability.

244 **Three renal processes**

Define filtration, reabsorption, secretion, and excretion and write the equation that relates them for any solute.

245 **Filtration membrane and selectivity**

Describe the three layers of the filtration membrane and predict which plasma components appear in the filtrate based on size and charge.

246 **Net filtration pressure and GFR**

Calculate net filtration pressure from glomerular hydrostatic pressure, capsular pressure, and colloid osmotic pressure and predict the effect of each change on glomerular filtration rate.

247 **Regulation of GFR**

Compare myogenic autoregulation, tubuloglomerular feedback, sympathetic control, and hormonal control and predict the effect of afferent and efferent arteriolar constriction on filtration rate.

248 **Renal clearance**

Calculate the clearance of a substance and use inulin and creatinine clearance to estimate glomerular filtration rate.

249 **Clearance and handling inference**

Use a clearance value relative to glomerular filtration rate to determine whether a substance is net reabsorbed or net secreted.

250 **Proximal tubule reabsorption**

Explain how the sodium gradient drives reabsorption of glucose, amino acids, and water in the proximal tubule and state why reabsorption there is described as isosmotic.

251 **Transport maximum and renal threshold**

Distinguish transport maximum from renal threshold and explain the appearance of glucose in urine when plasma glucose is elevated.

252 **Tubular secretion**

State the purpose of tubular secretion and name substances secreted including hydrogen ion, potassium, and organic anions such as drugs.

253 **Countercurrent mechanism**

Explain how the countercurrent multiplier of the loop of Henle and the countercurrent exchanger of the vasa recta build and preserve the medullary osmotic gradient.

254 **Antidiuretic hormone and urine concentration**

Trace the response to dehydration from osmoreceptor firing through antidiuretic hormone release to aquaporin insertion and concentrated urine.

255 **Renin angiotensin aldosterone system**

Trace the renin angiotensin aldosterone pathway from the stimulus for renin release to the effects of angiotensin II and aldosterone on vessels, tubules, and blood pressure.

256 **Natriuretic peptides**

Explain how atrial natriuretic peptide opposes the renin angiotensin aldosterone system and state the stimulus for its release.

257 **Potassium handling**

Explain how aldosterone and plasma potassium regulate potassium secretion in the collecting duct and predict the effect of a potassium wasting diuretic.

258 **Micturition**

Trace the micturition reflex and explain how the internal and external sphincters allow voluntary control.

259 **Urinalysis**

Perform a urinalysis measuring specific gravity, pH, glucose, protein, ketones, and blood and relate any abnormal finding to the renal process that failed.

260 **Renal function calculation**

Calculate glomerular filtration rate, clearance, and filtered load from a data set and interpret the values in a clinical case.

261 **Volume and osmolarity disturbances**

Classify a disturbance as volume depletion or overload and as hypoosmotic, isosmotic, or hyperosmotic and predict the resulting fluid shift between compartments.

AFTER THE EXAMS

Weeks 15 to 15

7 competencies. Not examined. Week 15 runs after Midterm 2 closes.

Week 15 The slow pH lever, and putting it all together

Mon Dec 14 to Wed Dec 16, three days · 7 competencies

262 **Renal control of pH**
Describe how the kidney secretes hydrogen ion and reclaims or generates bicarbonate and explain why renal compensation is slower and more complete than respiratory compensation.

263 **The four acid base disorders**
Classify a disorder as respiratory or metabolic and as acidosis or alkalosis from pH, partial pressure of carbon dioxide, and bicarbonate.

264 **Compensation**
Determine whether an acid base disorder is uncompensated, partially compensated, or fully compensated and name the system doing the compensating.

265 **Arterial blood gas interpretation**
Interpret a set of arterial blood gas values and match the result to a plausible clinical cause.

266 **Multisystem integration case**
Given a clinical scenario, trace the disturbance through at least three organ systems and identify the compensatory responses in the order they occur.

267 **Exercise as an integrated response**
Build an integrated account of moderate exercise across the muscular, cardiovascular, respiratory, endocrine, and renal systems.

268 **Drawing based synthesis check**
Draw and annotate a homeostatic pathway from memory and explain each step aloud or in writing without notes.

If one of these does not make sense yet

That is what the week is for. A competency is written as the finished skill, the thing you can do on the far side of the week, not the thing you already know walking in. Bring the ones that stay stubborn to my Virtual Office and we will take them apart.

Dr. Sharilyn Rennie